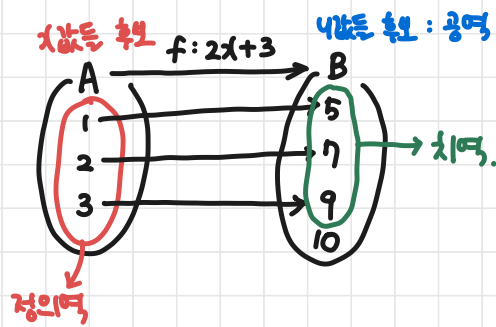
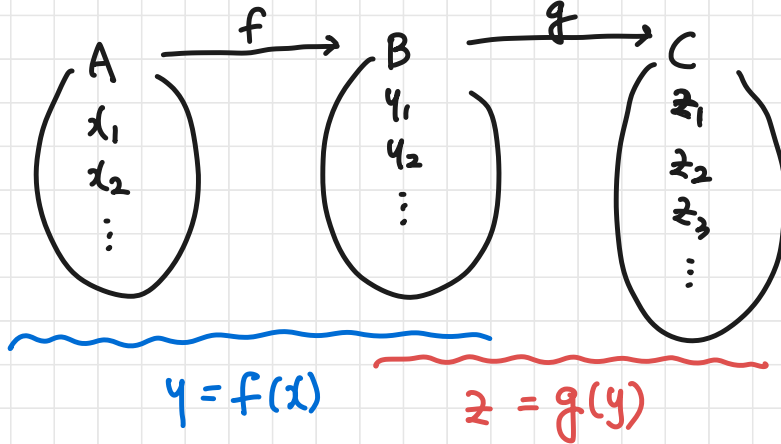


Recap) 함수 : $y = f(x)$

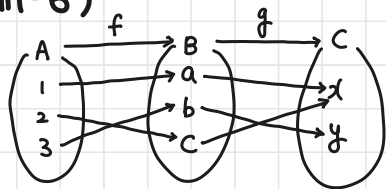


• 합성함수 : 한 함수의 출력이 다른 함수의 입력으로.



$$\begin{aligned} z &= g \circ f(x) = (g \circ f)(x) \\ &= g(f(x)) = g(y) \end{aligned}$$

#예1-6)



$$- (g \circ f)(1) = g(\underline{f(1)}) = g(a) = x$$

$$- (g \circ f)(2)$$

$$f(x) = -2x^2 + 5x + 1 \rightarrow \text{다항함수}$$

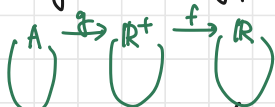
$$: x \in \mathbb{R}, y \in \mathbb{R}$$

$$g(x) = \sqrt{2x-1} : x \in 2x-1 \geq 0 \Leftrightarrow x \geq \frac{1}{2} \\ y \in \mathbb{R}^+$$

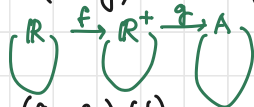
#예1-7)

$$f(x) = -2x^2 + 5x + 1, \quad g(x) = \sqrt{2x-1}$$

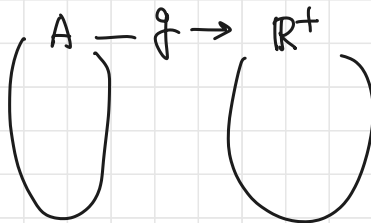
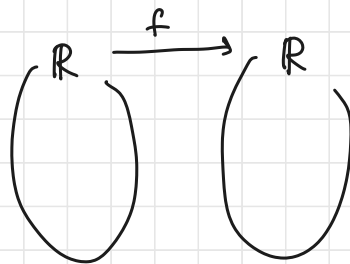
$$\bullet (g \circ f)(1) = g(f(1)) = g(-2+5+1) = g(4) = \sqrt{8-1} = \sqrt{7}$$



$$\bullet (f \circ g)(1) = f(g(1)) = f(1) = 4$$



$$\bullet (g \circ g)(1)$$



11-8)

$$f(x) = 2x - 1, \quad g(x) = -x + 1, \quad h(x) = -2x^2$$

$$\bullet \quad h \circ (\underbrace{g \circ f}_{\star})(x) = h(\star(x)) = h((g \circ f)(x)) \\ = h(g(f(x)))$$

$$\bullet \quad (\underbrace{h \circ g}_{\dot{g}}) \circ f(x) = \dot{g} \circ f(x) = \dot{g}(f(x)) = (h \circ g)(\underbrace{f(x)}_y) = (h \circ g)(y) \\ = h(g(y)) = h(g(f(x)))$$

$$\begin{aligned} h(g \circ f)(x) &= h(g(f(x))) \\ &= h(g(\underbrace{2x-1})) = h(- (2x-1) + 1) = h(-2x+2) \\ &= -2 \cdot (-2x+2)^2 = -2(4x^2 - 8x + 4) \\ &= -2 \cdot 4(x^2 - 2x + 1) = -8(x-1)^2 \end{aligned}$$

$$g \circ f(x), f \circ g(x) \rightarrow \text{함수성함수}$$

$$f \circ f(x), g \circ g(x), \underbrace{f \circ f \circ f \circ f(x)}_{4회} = f^4(x)$$

#예1-9) $f(x) = x + k$

$$f^2(x) = f \circ f(x) = f(f(x)) = (x+k) + k = x + 2k$$

$$f^3(x) = f(f^2(x)) = f(x+2k) = (x+2k) + k = x + 3k$$

$$f^4(x) = f(f^3(x)) = f(x+3k) = (x+3k) + k = x + 4k$$

$$f^4(3) = 15 = 3 + 4k$$

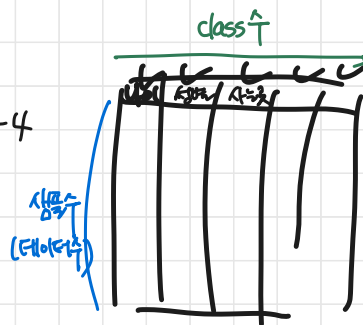
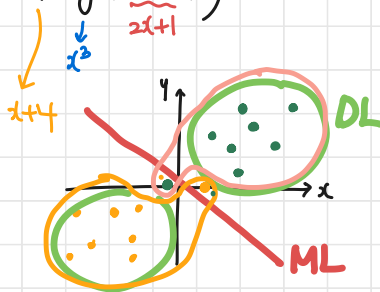
$$4k = 12 \quad \therefore k = 3$$

ML $\rightarrow$ 레이어 1

DL $\rightarrow$ 레이어 99

x $\xrightarrow{f}$ y
 예제
 $f(x) = w_3 x^3 + w_2 x^2 + w_1 x + b_1$
 $w_1 x^2 + w_2 x + b_1$
 x $\xrightarrow{w_1 x + b_1 : 2x+1}$ y
 3

$$f(g(h(x))) = (2x+1)^3 + 4$$



• 항등함수 : $y = x$

: $y = f(x)$ 에서 $x = y$ 면, f 는 항등함수.

$$I: A \rightarrow A \quad I(x) = x$$

f 가 항등함수 이면 $y = f(x) = I(x) = x$

